

a What each validation target can establish

● Validation target	✓ Strongest claim it supports	✗ Does not establish on its own
Known motion (phantom, STRAUS, MRXCAT2.0)	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan	precision, not accuracy	bias
LGE or tissue characterisation	biological association, not motion error	pointwise displacement error
Outcome	prognostic association, not clinical utility	decision impact

b A staged programme for regional or focal claims

✓ **Pre-specify the measurement contract**
 estimand · reference · tolerance · data splits

Resources (Table 4)

1 Stage 1 · Known motion
 vary width, extent, amplitude, timing, resolution and noise

STRAUS: 3 templates × 6 states

MRXCAT2.0: programmable generator

CMAC dynamic phantom

2 Stage 2 · Paired material reference
 cine with DENSE or tagging in one session; registered; reference blinded

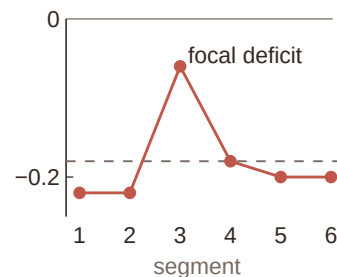
Stanford: 51/55 · not file-verified

CMAC: 15 volunteers, 12 landmarks

3 Stage 3 · Tissue and decisions
 LGE, responsiveness, decision impact

Clinical cohorts with LGE or outcomes

c Equal-mean test



Uniform (dashed) and focal (solid) fields share a global mean of -0.18 . Recovering the mean is not a pass.